Accessibility Analysis of the Bosnia and Herzegovina Universities' Websites in the Relation to Their Position on the Webometrics Ranking List

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Abstract - **In today's digital age, presence on the Internet and the Web mean presence in general life. This is generally more pronounced in the context of higher education because these institutions are the primary drivers of overall social development. The quality and highly accessible website of each university is a must as these websites are generally the first contact and first impression for new potential students and also the first step for establishing cooperation with the social economy and society. For this reason, websites need to meet some of the basic standards when it comes to accessibility of the content they provide and it is essential to ensure that content on the websites can be accessed by people with disabilities. This paper deals with the analysis of websites of universities in Bosnia and Herzegovina in the context of their accessibility, from one side, and their position on the international scene in the context of the Webometrics list of universities, on the other side. Based on the obtained results and conducted analysis, recommendations were given on how to improve some of the elements on websites in order to make these websites more accessible in accordance with the standards related to this topic.**

Keywords - web accessibility; accessibility issues; web content accessibility guidelines; higher education; webometrics;

I. INTRODUCTION

Today digital communications and especially internet play an important part in people's everyday life. Web sites as major sources of information are becoming irreplaceable in every sphere of human life, from shopping, promotions, entertainment to education which in these circumstances becomes very close to everyone. Because of this fact, it matters to create accessible websites for all kind of users. Especially, web site usage problems becomes very significantly expressed for website users with disabilities. Understanding these kind of users is the first step and right way to understand the need for everyone [1].

In accordance with European Disability Strategy 2010-2020 [2] one of six people in Europe has a severe to mild disability that affects opportunities of those people to participate in social activities and the economy. On the other hand, the process of biological aging of the population in any country leads to an increased number of people who have a problem with some kind of disabilities [3]. This leads to increasing the total number of people who need websites that are adapted to this category of people.

Digital Agenda for Europe emphasis web accessibility as a very important objective to strengthen and to provide full implementation of this objective. European directive has been issued in which states that all public websites should be accessible starting from September 2020 [3] [4].

There are millions of websites in the world that serve for different purposes and the number of educational sites, especially ones in higher education, is not smaller in number in that context. Website to achieve its full popularity and success must be accessible in every way, in different search engines and in different type of devices in such way that it can support assistive technologies [5]. However, not all websites comply with standards and regulative proposed by the World Wide Web consortium (W3C) [6].

Website compliance with accessibility standards that are used both for people with and without disabilities in accordance with published guideline by the W3C is known as Web Content Accessibility Guidelines (WCAG) [7][8]. In year 1999 the first version of WCAG was published while the second version, WCAG 2.0 was published in 2008. In the year 2012, WCAG 2.0 was approved as ISO standard IOS/IEC 40500 [9].

WCAG is based on four essential principles: perceivable, operable, understandable and robust.

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Perceivable considers the three main senses needed to perceive web content such as sight hearing and touch. Operable principle defines various methods for web sites accessibility such as navigation alternatives and appropriate user interfaces for people with disabilities. Understandable principle defines the forms of correct interpretation of the content. Robust principle considers compatibility with both current and future technologies.

These principles are supported by guidelines which include several success criteria and which in return are pathways on how to achieve these principles. The guidelines include 61 success criteria organized according to three levels of conformity [9] [10]:

Level A, the minimum level, which includes 25 success criteria.

Level AA, the middle level, in which a website must meet all the criteria for both levels A and AA. It includes 13 additional success criteria.

Level AAA, the highest level. At the AAA level, a website must comply with all the criteria in levels A, AA and AAA. It includes 23 additional success criteria.

For the public websites in Europe, the AA level of conformance is required [3].

Universities play a very important role in a community and their websites must be accessible to every person, especially as they are an important communication channel between students, teachers, and administrative staff and at the end of all community. Users' first impression of the educational institution comes from the first contact with the institution, and in this digital age, universities website is the starting point.

The main purpose of this study was to investigate whether universities websites in Bosnia and Herzegovina meet the current accessibility guidelines, their compliance with WCAG 2.0 guidelines in relation to their ranking on the webometrics list [11] (July 2020. edition published by the Webometrics). Webometrics list, in its context, have some advantages and disadvantages [12], but it is still the ranking list that most of the world universities refers to. This study also provides some recommendations and guidelines for website developers on how to improve their website accessibility.

II. RELATED WORK

There have been several studies conducted with the aim to evaluate public websites accessibility in higher education in the world but as far as we know none of the studies exists in the context of website accessibility and their position on the webometrics list in Bosnia and Herzegovina.

Acosta-Vargas et al. [13] presented a study on the assessment process of websites accessibility policies that were adopted by 51 university websites across the world which were ranked by Webometrics. The web accessibility assessment was aimed at validating compliance with the WCAG 2.0. The primary goal of their study was to identify a group of universities with an academic prestige and to determine their level of commitment on providing accessible information about university facts and activities presented on their main websites as well as making it available to every user including people with disabilities. The results presented in study confirmed that 44 out of 51 universities that were evaluated did not reached the acceptable level of accessibility policies.

Acosta-Vargas et al. [10] presented web accessibility problems that were identified for websites of 348 universities located in Latin America and in accordance with their position on the Webometrics list. Obtained results suggests that these websites very often have problems that are primarily related to the missing of alternative text for images. Authors also noted that these websites don't fulfill web accessibility requirements based on the WCAG 2.0.

Abid Ismail et al. [14] presented study of an accessibility analysis for 59 higher education websites of Portugal. Both Polytechnic and University websites were incorporated into the study. The accessibility analysis was carried out with three analysis tools: AChecker [15], WAVE [16], and aXe [16]. Based on the obtained results, outliers of accessibility violations were calculated with respect to websites of Polytechnic Institutes and Universities of Portugal. Moreover, authors also suggested and discussed the important clues on how to solve the accessibility violations of websites along with their issue descriptions. Among the selected websites in this study their analysis report provided feedback to the website administrators and web developers on how to improve the accessibility evaluation status of these websites.

In Acosta-Vargas et al. [5] study, research had couple of purposes: 1) to suggest a model for accessibility assessment in accordance with WCAG 2.1 standard; 2) to correlate Webometrics and SCImago rankings in order to improve visibility and 3) to determine whether an accessible website can improve visibility. As a case study, the authors applied analysis to 30 websites from Latin American universities. The results indicated that although an educational institution is in the first ranking of Webometrics and SCImago classification, they do not necessarily satisfy web accessibility requirements.

K. Kous et al. [17] in their study included an analysis of the faculties websites accessibility between 2018 and 2019. The objective of their research was to investigate the WCAG 2.0 compliance level of faculties' websites. Obtained results indicated that none of the assessed websites reached accessibility level defined within WCAG 2.0.

Rita Ismailova and Yavuz Inal [18] in their study aimed to test the websites of the highly rated universities in Kyrgyzstan, Kazakhstan, Azerbaijan and Turkey. Presented results showed that university websites are much more popular in Turkey. Website developers at Turkish universities pay more attention to the website performance followed by websites of Azerbaijani, Kyrgyz and Kazakh universities. Significant number of university websites that were included in the study did not meet the WCAG 2.0 accessibility criteria.

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W Arasid et al. [19] in their study analyzed 13 universities' websites in West Java, Indonesia by using TAW as an evaluation tool. Website of State Universities in West Java was not fully qualified in context of website accessibility because it violates some criteria from WCAG 2.0 guidelines in terms of website accessibility.

III. METHODS AND TOOLS

This study was conducted on a sample of 45 universities in Bosnia and Herzegovina according to the Webometrics list. At the time of writing this paper (11.01.2021.), there are a total of 48 higher education institutions (HEIs) on the Webometrics list, however, by analyzing the data and availability of websites, 3 websites were not available for analysis and therefore were not taken into consideration and further content analysis. These three universities are Kallos Tuzla (9th position on Webometrics list), American university (23rd on Webometrics list) and ITC Interlogos Center Kiseljak (47th on Webometrics list).

Data were taken from two sources. The first source was the website itself available via URL, while the second part of the data was taken from the official Webometrics website from the part related to universities in Bosnia and Herzegovina. The last report published in July 2020, Edition 2020.2.4, was observed [20].

In addition to the ordinal number, which is also the number of position of University on the Webometrics list, with the focus on Bosnia and Herzegovina (B&H), we have added an acronym for each University for easier tracking and analysis.

The tool used to analyze website accessibility in terms of compliance with the WCAG 2.0 standard is the TAW [21] software. TAW is an online analytics tool that supports HTML, CSS and JavaScript analysis. The TAW tool, in addition to the AChecker [15] and TotalValidator [22] tools, has been also used because it is more suitable for large-scale evaluations and also provides a compact but still detailed report on individual WCAG criteria [3].

For the analysis of accessibility of universities web sites in this paper, four steps were applied, which are presented below:

Step 1: Data were collected for all 45 universities from the Webometrics page for Bosnia and Herzegovina. It can be noticed that the Webometrics list, edition for July 2020, Edition 2020.2.4, contains a larger number of universities and other HEIs than the one shown in the Agency for Development of Higher Education and Quality Assurance of Bosnia and Herzegovina – HEA [23] register, ie, not all universities shown on the webometrics list are accredited ones in BiH.

Step 2: Data related to the subject universities are filtered and stored in spreadsheets (Microsoft Excel). The sample data for the work was selected based on the availability of websites for analysis.

Step 3: Using the TAW tool, the analysis of the homepage of each of the universities was performed. After running the TAW tool for each of the websites, the same returned several parameters as a result. These parameters are: error, perceivable, operable, understandable, and robust.

Step 4: Analysis of results. In this step, all the obtained results were analyzed in terms of compliance with the WCAG standard.

After completing steps 1, 2, and 3, data related to the accessibilities parameters returned by the TAW tool for each website were collected and these results are shown in Table I.

TABLE I. RESULTS OF ANALYZED WEBSITES RETURNED BY THE TAW TOOL.

No	University	Acronym	E	P	O	U	R
1	University of Sarajevo	UNSA	79	30	1	0	48
2	University of Tuzla	UNTZ	188	54	1	5	128
3	International University of Sarajevo	IUS	137	51	64	1	21
4	International BURCH University Sarajevo	IBU	181	58	91	1	31
5	University of Banja Luka	UNIBL	92	7	78	1	6
6	University of Mostar	SUM	443	14	15	2	412
7	University of East Sarajevo	UES	119	28	84	1	6
8	Sarajevo School of Science & Technology	SSST	16	8	6	0	2
10	European University Brčko	EUBD	110	32	54	4	20
11	University of Bihać	UNBI	36	7	20	3	6
12	University Dzemal Bijedic Mostar	UNMO	37	11	6	2	18
13	International University of Travnik	IUT	128	49	33	3	43
14	University of Zenica	UNZE	25	1	13	0	11
15	Paneuropean Apeiron University Banja Luka	APEIRON	44	9	31	1	3
16	University Vitez in Travnik	UNVI	60	20	21	5	14
17	University of Business Studies Banja Luka	UPSBL	52	10	27	2	13
18	Slobomir P University	SPU	51	12	18	3	18
19	Banja Luka College	BLC	85	31	18	7	29
20	University Sinergija	SIN	136	11	6	2	117
21	University of Travnik	UNT	76	0	0	2	74
22	IPI Academy Tuzla	IPI	72	38	13	1	20
24	Herzegovina University	HERCUNI	83	33	23	4	23
25	PIM University Banja Luka	PIM	96	31	26	5	34
26	Banja Luka College of Communications	KFBL	54	26	19	0	9
27	Independent University of Banja Luka	NUBL	51	15	19	4	13
28	ITEP Higher School	ITEP	113	56	13	19	25
29	Sarajevo Film Academy	SFA	20	3	17	0	0

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30	University College CEPS Kiseljak	CEPS	165	32	9	2	122
31	College of Computer Science and Business Communications eMPIRICA	EMPI	52	36	6	2	8
32	High Medical School Prijedor	VMSPD	28	14	0	6	8
33	Higher School Logos Center Mostar	LOGOS	158	29	11	2	116
34	International University of Gorazde	IUG	92	34	16	14	28
35	University of Modern Sciences CKM	CKM	39	6	21	1	11
36	High College for Applied and Law Sciences PROMETEJ	PROBL	14	5	3	1	5
37	Business and Technical College Doboj	VPTS	155	22	25	1	107
38	Sokolac College of Service Business	VUB	316	43	64	4	205
39	Higher School of Finance and Accounting FINra Tuzla	FINRA	21	7	2	2	10
40	University of Travnik, Faculty of Business Economics	FMPE	60	17	32	2	9
41	University of Bijeljina	UBN	35	8	21	1	5
42	PI Veterinary Research Institute	VIRS	1	0	0	1	0
43	College for Tourism and Management Konjic	VSTIM	47	6	22	5	14
44	College of Economics and Informatics Prijedor	VSEIP	124	48	16	24	36
45	Higher School of Hotel Management and Tourism Trebinje	THT	5	1	0	1	3
46	Cazin International High School	VMS	21	3	5	0	13
48	Faculty of Cosmetology and Esthetic	FCE	113	13	18	1	81

E – Error; P – Perceivable; O – Operable; U – Understandable; R – Robust;

IV. RESULTS ANALYSIS

The relationship between the ranking of universities according to the Webometrics list and the number of accessibility errors generated by the TAW tool is presented on Fig. 1.

Analyzing the score for 45 university websites, the correlation between university rank according to Webometrics and accessibility error was -0.2329 which is a fairly strong negative correlation. This tells us that the websites of highly ranked universities do not necessarily have an accessible website, and a similar conclusion was drawn in the Acosta-Vargas [10] study. If we look at the first seven universities on the scale (Fig. 1), we can notice that these universities have the most pronounced errors when it comes to web site content accessibility, which is very interesting if we take into account that these universities should be among the first in the country to meet all standards when we talk about accessibility of their websites.

Taking into account the fact that each university website is created and maintained using some kind of a content management system (CMS, Content Management System), we conducted an analysis on how used CMS influences on the presence of the number of accessibility errors obtained using the TAW tool.

The total accessibility error for the observed universities ranked on the Webometrics list was 4030. The university websites used different CMS systems for their needs. The most common CMS is WordPress [24], followed by some custom made solutions and to a lesser extent Joomla [25] and Drupal [26].

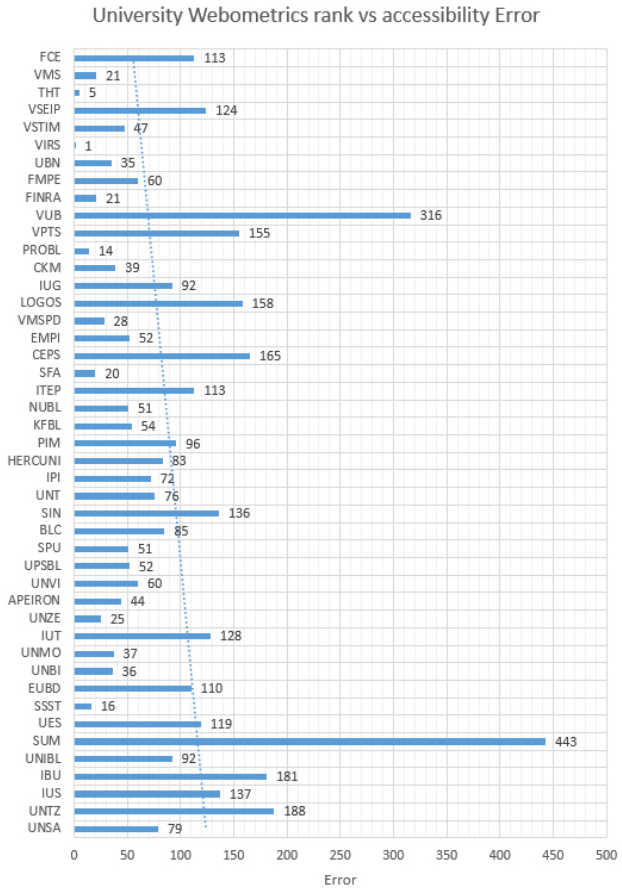


Figure 1. The relationship between university ranking and the number of accessibility errors generated by the TAW tool.

The fact that WordPress is the most used is not surprise at all if we look at data from the *BuiltWith* list [27], where it is stated that WordPress has a 41% share on the entire web when it comes to websites.

If we look at the participation of each of the CMS systems in the total accessibility error, with the respect of the popularity of use, websites with the WordPress system have practically the largest participation in the total accessibility error of the observed university websites as it is shown in Fig. 2.

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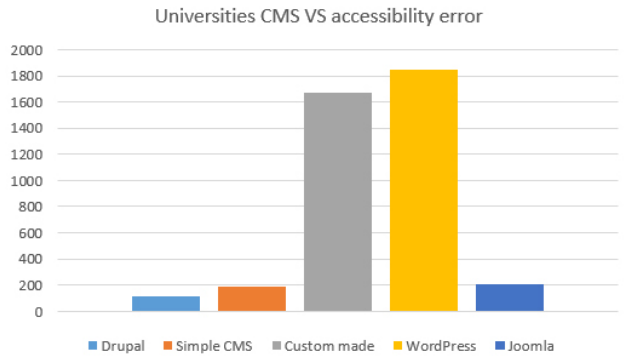


Figure 2. CMS systems that carry the greatest impact in overall accessibility error.

This fact related to WordPress leads us to one possible conclusion that most universities, when they set up their website with a ready-made system, such as for example WordPress, do not further work on its optimization and customization to reduce the accessibility error of the website itself.

For this reason, the system administrator or creators of universities websites should do an accessibility check of their websites, no matter which one CMS they use. If we are talking specifically about WordPress, which is the most represented in this study, then website creators should install some additional plugins that would do accessibility check of their websites [28] [29] [30].

Table II provides a detailed overview of the accessibility error participation in the total accessibility error and number of CMSs used for all analyzed universities.

TABLE II. REVIEW OF ALL PARAMETERS RELATED TO THE PARTICIPATION OF AN INDIVIDUAL CMS IN AN INDIVIDUAL AND TOTAL ERROR.

CMS	CMS count	Accessible errors	% in total error
Drupal	2	114	2,83
Simple CMS	1	188	4,67
Custom made	15	1668	41,39
WordPress	23	1851	45,93
Joomla	4	209	5,19
TOTAL	45	4030	

After the analysis, we can observe that out of 45 ranked universities, the percentage of errors (their share) for perceivable, operable, understandable, and robust in the total error (4030 errors) of all universities is as follows: perceivable 24.04%, operable 24.52%, understandable 3.67%, and robust 47.77 %. This significant share in the total error when it comes to robustness is influenced by two universities, University of Mostar - SUM (ranked as a 6th) and the College of Service Business Sokolac - VUB (ranked as a 38th). Essentially, we have the biggest accessibility error with the perceivable and operable components. If we look in a little more detail in these two criteria, ie what are the elements on the web pages that make the biggest problem in terms of accessibility, they are mostly images with empty or without alt attributes, or images that requires long description, no marking of input text labels, possible elements on web sites that do not conform with responsive design and thus when resizing pages, some content can be hidden, problems with empty links or links with same text that points to different locations. When it comes to robustness, two previously mentioned universities (SUM and VUB) could generate significant problems for users when accessing these web sites because it is possible that web content on these websites is not compatible with a variety of "user agents", like web browsers, assistive technologies, and other means of accessing web content.

Fig. 3 shows principles of web accessibility, error participation of each principle (perceivable, operable, understandable and robust) in cumulative error for each analyzed university. We can see many variations but still we have very big cumulative accessibility error for top ranked universities.

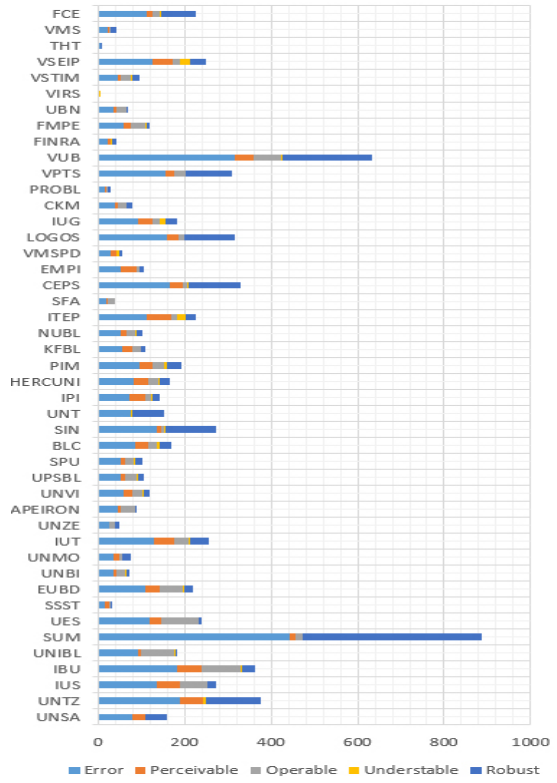


Figure 3. Universities ranked by the Webometrics and total, perceivable, operable, understandable, and robust errors

V. CONCLUSION

In this paper, for practical and experimental reasons, the WCAG 2.0 standard was used, which was approved as an ISO/IEC 40500:2012 standard, and not WCAG 2.1, which was published in 2018 as a W3C recommendation. Future research will be focused on the analysis of the fulfillment of the criteria from WCAG 2.1 at HEIs in Bosnia and Herzegovina.

After a detailed analysis of the university website, we can conclude that most of them have a large number of accessibility errors. Universities that are highly positioned on the Webometrics list or that should essentially have a good academic reputation essentially

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do not meet an acceptable accessibility level and are practically one of the leaders when it comes to not providing adequate conditions for displaying accessible information to every kind of user, regardless whether that user has any disabilities or not.

The study revealed a large number of accessibility errors in the perceivable and operable part of the WCAG standard on universities websites and mostly it was about images with empty or without alt attributes, or images that require long description, no marking of input text labels, possible elements on web sites that do not conform with responsive design and thus when resizing pages some content can be hidden, problems with empty links or links with same text that points to different locations. In addition, there are a large number of warnings on the created web sites content in accordance with accessibility guidelines. Although warning does not have the same power as error, their correction would likely lead to a reduction of possible accessibility errors.

Since the content on web pages is mostly created using some CMS tools, it is necessary to include, in addition to a human expert for error estimation and correction, and add-ons that some CMS systems may have, whose main purpose is to automatically adjust elements or web content on web pages to achieve higher degrees of accessibility. In this way, developers and designers of university websites are encouraged to apply the WCAG standard in their work as a starting point when developing accessible websites.

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